Saatvik Palli

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Education

University of Illinois Urbana-Champaign — Champaign, IL
Master of Computer Science — *Aug. 2026 – Dec. 2027*

Michigan State University — East Lansing, MI
B.S. in Computer Science, GPA: 3.9/4.0 — *Aug. 2022 – May 2026*

Experience

Founding Engineer — Aug. 2025 – Present
Nexxup Music Discovery — *East Lansing, MI*

- Enabled week-over-week momentum ranking for **144 independent artists** by building a daily ETL pipeline accumulating 380,000+ datapoints across 10 streaming and social platforms.
- Cut artist leaderboard load times from **3-4 seconds to ∼600 ms** by shrinking the initial bundle 57%, replacing $O(n^2)$ scans with $O(1)$ hash-set lookups, and deduplicating concurrent API requests behind a TTL cache.
- Defined the data model and service boundaries while mentoring **3 developers and 2 designers** over 9 months through branch-per-developer workflow, code review, and 110+ PRs.

Data Analytics & Engineering Intern — May 2024 – Aug. 2026
Auto-Owners Insurance — *Lansing, MI*

- Spearheaded upgrading licensing management to an internal SQL Server system, saving **$115K over 5 years**, by engineering ETL pipelines and stored procedures that surfaced real-time compliance status.
- Developed a neural network capable of scoring **10,000+ claims per second**, producing real-time difficulty ratings used to balance caseloads across adjusters, with scikit-learn and MLflow.
- Built a Power BI scoring model ranking **500+ adjusters across 43 branches** on **8 weighted claim-handling metrics**, exposing systematic gaps in rep training and technology adoption.
- Analyzed thousands of claims with Pandas and NumPy to project 18-month cost savings of chatbot-assisted salvage handling, generating reports that strengthened vendor negotiations and contract renewal.

Software Developer — Aug. 2025 – Dec. 2025
Magna International, Michigan State Capstone — *East Lansing, MI*

- Developed a CAD classification pipeline enabling engineers to find functionally similar parts across **2,000+ automotive components**, by tagging 3D meshes with a vision LLM using extracted geometry statistics.
- **Increased tagging accuracy by 64%** through prompt A/B testing and multi-view rendering optimization.
- **Cut per-request API cost by ∼50%** and eliminated rate-limit failures by moving tagging to an OpenAI Batch API pipeline that chunked 1,800 parts into 90 asynchronous jobs with state persistence for interrupted runs.

Executive Board — Aug. 2024 – May 2026
Imagine Software Club at MSU — *East Lansing, MI*

- Interviewed and matched **∼50 students to 6 startup projects**, aligning technical experience to effective teams.
- Designed and hosted a Bash scripting workshop for **60+ students** by hosting a custom server environment.

Projects

Half Full Calendar | *JavaScript, Electron, Firebase, Google Calendar API*

- Built two-way Google Calendar synchronization backed by Firebase persistence, enabling offline access and cross-session state for a desktop calendar since released on the **Mac App Store and Microsoft Store**.
- Contributed frontend fixes to event creation and viewing flows on a team-built Electron application.

Insurance Risk Factor Neural Network | *Python, Flask, Scikit-Learn, Plotly, Git/GitHub*

- Led a team of 3 to build a web app visualizing health risk factors and projected financial loss.
- **Improved model accuracy by 8%** on insurance dataset by tuning a scikit-learn MLP.

Technical Skills

Languages: Python, SQL, Bash, JavaScript, C++, C, DAX, HTML/CSS
Technologies: SQL Server, GCP, Docker, React, Flask, Firebase/Firestore, Pandas, NumPy, PyTorch, Scikit-learn, MLflow, Power BI, Plotly, Git/GitHub, Electron
Concepts: Data Modeling, ETL, Data Warehousing, Distributed Systems, Database Systems (SQL & NoSQL), Query Optimization, Algorithms & Data Structures, Machine Learning, Linux
Interests: Jazz trumpet, Eagle Scout, Brazilian jiu-jitsu, poker